5.4 Equilateral and Isosceles Triangles

Name _____ Period ____ Date _____

Geometry 5.4 (30-minute plan) · Student sheet

I can…

- use the Base Angles Theorem to find angle measures.
- use the converse to find side lengths.
- use the properties of equilateral triangles.

Vocabulary · fill in the missing word

Legs (of an isosceles triangle) — the two _____ sides

Vertex angle — the angle _____ by the legs

Base — the side _____ the vertex angle

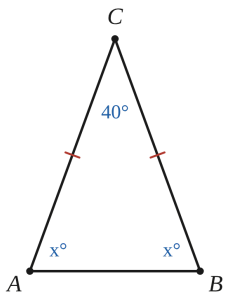
Base angles — the two angles _____ to the base

Part 1 · Notes and examples (fill in each note, then work the example)

Base Angles Theorem: if two sides of a triangle are congruent, then the angles _____ them are congruent.

Example 1

An isosceles triangle has a vertex angle of 40°. Find each base angle.



Converse of the Base Angles Theorem: if two angles of a triangle are congruent, then the sides opposite them are _____.

Example 2

Two angles of a triangle measure 50°, and the sides opposite them measure (3x − 2) and (x + 10). Find x and the length of each side.

Corollaries: a triangle is equilateral if and only if it is _____, so each angle of an equilateral triangle measures 60°.

Example 3

An equilateral triangle has sides of (4x − 3) and (2x + 9). Find x, the side length, and each angle.

Part 2 · Partner practice

Partner practice

An equilateral triangle has sides of (5x − 8) and (3x + 4). Find x and the side length.